MATH 351 Numerical Methods with Applications
Fall 2026
Group Project Description

We have data on the population of Kazakhstan for several years. The number of people corresponds to the beginning of each year, i.e., January 1.

Year	2021	2022	2023	2024	2025	2026
All population	18,879,552	19,503,159	19,766,807	20,033,842	20,286,084	20,499,822
Urban population	11,151,376	11,991,238	12,209,896	12,451,192	12,774,743	13,085,110
Rural population	7,728,176	7,511,921	7,556,911	7,582,650	7,511,341	7,414,712

For the computational tasks below, MATLAB must be used. For the computational tasks, use and adapt the MATLAB codes provided in the lecture slides. Ensure that your group understands the algorithms and clearly explains any modifications made for this project. Provide the MATLAB code used to implement all numerical methods required in this project. Include relevant code excerpts in the report to explain your implementation, and submit the complete, runnable code separately as .m files.

1. **Interpolation and prediction.**

   Utilize the population data from 2021 to 2025 to estimate the values of all population, urban population, and rural population for 2026.

   Consider the following two models:

   - **Model 1:** Lagrange interpolation polynomial;
   - **Model 2:** Cubic spline interpolation.

   For each of the three population categories:

   (a) construct both numerical models;
   (b) use each model to predict the corresponding population in 2026;
   (c) compare the predicted values with the actual 2026 values given in the table;
   (d) plot the available data together with both numerical models on the same figure; For each population category, plot both numerical models over 2021–2026 on the same figure. Mark the 2021–2025 data points with circles and the actual 2026 population with a star. Use distinct line styles for the two models and include axis labels and a legend.

2. **Model validation and error analysis.**

   A model should not be evaluated using only one prediction. Therefore, perform the following backtesting experiments for **both Model 1 and Model 2**:

   $$\begin{aligned} &\text{use data from 2021–2023} \longrightarrow \text{predict 2024},\\ &\text{use data from 2021–2024} \longrightarrow \text{predict 2025},\\ &\text{use data from 2021–2025} \longrightarrow \text{predict 2026}. \end{aligned}$$

   Perform these experiments separately for

   - all population,
   - urban population,
   - rural population.

For every prediction, calculate the absolute error

$$E_{\text{abs}} = |P_{\text{actual}} - P_{\text{predicted}}|$$

and the relative error

$$E_{\text{rel}} = \frac{|P_{\text{actual}} - P_{\text{predicted}}|}{|P_{\text{actual}}|}.$$

Present the results in clearly organized tables.

Based on all the back-testing results, answer the following questions:

(a) Which model gives the smaller prediction errors?
(b) Is the same model consistently better for all population categories and all prediction years?
(c) Is it reasonable to choose a numerical model based only on its prediction for one year? Explain.

3. **Sensitivity and stability of the numerical models.**

Real-world data may contain measurement, reporting, or rounding errors. Investigate how sensitive the two numerical models are to small changes in the input data.

Perform the following three numerical experiments:

(a) For the **all population** data, increase the 2024 value by 0.5%.
(b) For the **urban population** data, decrease the 2024 value by 0.5%.
(c) For the **rural population** data, increase the 2024 value by 0.5%.

For each experiment:

(i) reconstruct Model 1 and Model 2;
(ii) calculate the 2026 prediction before and after the perturbation;
(iii) calculate

$$S = \frac{|P_{\text{perturbed}} - P_{\text{original}}|}{|P_{\text{original}}|};$$

(iv) compare the sensitivity of the Lagrange and cubic spline models.

Based on your numerical experiments, discuss:

- which model appears to be more stable with respect to small changes in the data;
- whether a model that interpolates all given data points exactly must necessarily provide a reliable prediction;
- which of the two models you would select for the remaining tasks, and why.

4. **Population threshold and root-finding methods.**

Using the population data from 2023 to 2026 and the interpolation model that your group selected in Task 3, determine approximately when the total population of Kazakhstan will reach

$$20.7 \text{ million}.$$

Let $P(t)$ denote your selected population model. Formulate the problem as the nonlinear equation

$$F(t) = P(t) - 20,700,000 = 0.$$

Solve this equation numerically using the following methods:

(a) Bisection method;

(b) Secant method;

(c) Newton's method.

For each method:

- clearly state the initial interval or initial approximation(s);
- use the same stopping tolerance for all three methods;
- provide an iteration table containing the approximations and the corresponding residuals;
- report the number of iterations required;
- report the predicted year and month when the population reaches 20.7 million.

Compare the three root-finding methods in terms of convergence, computational effort, and reliability.

Do the three methods produce essentially the same final answer? If so, does this imply that the three numerical methods are equally efficient? Explain.

5. **Alternative methods for population prediction.**

   Are there alternative methods besides polynomial and cubic spline interpolation in the literature that could be explored for population prediction?

   Find and discuss at least **two peer-reviewed journal papers** related to population prediction or forecasting.

   Your discussion should:

   - identify at least two alternative modeling approaches;
   - briefly explain the main idea behind these approaches;
   - compare them with the interpolation methods used in this project;
   - discuss their possible advantages and disadvantages for population prediction;
   - identify your preferable model and explain your choice.

   Your discussion must refer to numerical evidence obtained in Tasks 1–4.

   Minimum length: 300 words.

6. **Rural population: prediction and interpretation.**

   Using the numerical model selected by your group in Task 3, and using all available data from 2021 to 2026, project the rural population of Kazakhstan in

   $$2028 \qquad \text{and} \qquad 2030.$$

   Discuss the following:

   - What behavior or trend do you observe in the rural population?
   - How reliable do you think the predictions for 2028 and 2030 are?
   - What are the main limitations of using your numerical model for long-term extrapolation?
   - If you were an investor or decision-maker, what actions or investments could potentially influence the future development of rural areas?

   Your discussion must use specific numerical results obtained in this project, including at least one prediction and one result from the error or sensitivity analysis.

   Minimum length: 200 words.

7. **Urban population: prediction and interpretation.**

   Using the numerical model selected by your group in Task 3, and using all available data from 2021 to 2026, project the urban population of Kazakhstan in

   $$2028 \qquad \text{and} \qquad 2030.$$

   Discuss the following:

   - What behavior or trend do you observe in the urban population?
   - How reliable do you think the predictions for 2028 and 2030 are?
   - What are the main limitations of using your numerical model for long-term extrapolation?
   - What opportunities and challenges could continued urban population growth create?
   - If you were an investor or decision-maker, which areas would you prioritize and why?

   Your discussion must use specific numerical results obtained in this project, including at least one prediction and one result from the error or sensitivity analysis.

   Minimum length: 200 words.

Report Requirements

The project must be completed by a team of a maximum of **three students**.

The report must be typed using Word with MathType or LaTeX (e.g., Overleaf) and should contain:

- a brief introduction;
- mathematical descriptions of the numerical methods used;
- all required numerical results;
- clearly labeled tables and figures;
- discussion and interpretation of the numerical results;
- MATLAB code; Include relevant MATLAB code excerpts in the report to explain your implementation. Submit the complete, runnable MATLAB code separately as .m files, including all scripts, functions, and data needed to reproduce the reported results.
- references used in Task 5;
- a short conclusion.

All figures must contain appropriate axis labels, legends, and captions. All numerical results should be reported with a reasonable number of significant digits.

All group members are responsible for the correctness and understanding of the complete submitted project.